



Lecture 2.1: Overhead Transmission Lines

EEE 321: Power System I

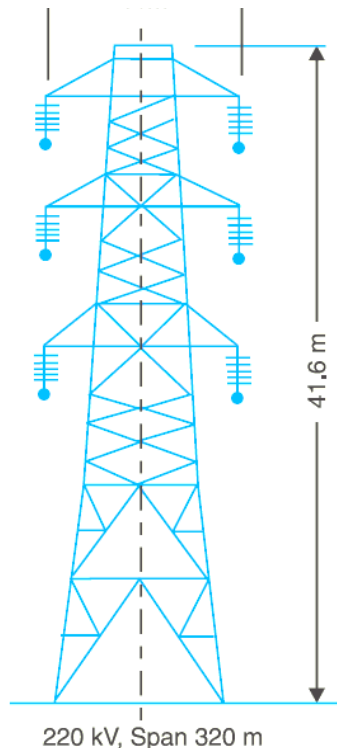
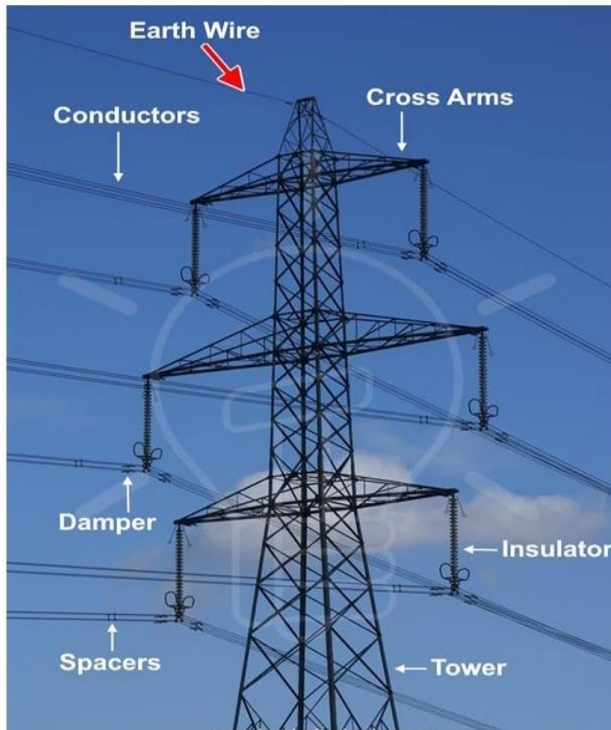
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Presentation Outline

- ❑ Main Components of Overhead Transmission Lines
- ❑ Conductor Materials
- ❑ Line Supports
- ❑ Insulators

Main Components of Overhead Transmission Line

- **Conductors** (Carry electric power)
- **Supports** (Poles or Towers)
- **Insulators** (Insulate the conductors from the ground)
- **Cross arms** (Provide support to insulator)
- **Miscellaneous items** (danger plates, lightning arrestors, anti-climbing wires, spacer, damper etc.)



Electric power can be transmitted or distributed either by means of underground cables or by overhead lines. The underground cables are rarely used for power transmission due to two main reasons.

1. Power is generally transmitted over long distances to load centres. Obviously, the installation costs for underground transmission will be heavy.
2. Electric power has to be transmitted at high voltages for economic reasons. It is very difficult to provide proper insulation to the cables to withstand such higher pressures.

Conductor Materials

Properties of Conductor Materials:

- **High electric conductivity;**
- **High tensile strength** in order to withstand mechanical strength;
- **Low specific gravity** so that weight per unit volume is small;
- **Low cost** so that it could be used for long distances;

All above requirements are not found in a single material. Therefore, while selecting a conductor material for a particular case, a compromise is made between the cost and the required electrical and mechanical properties.

Commonly Used Conductor Materials:

The most commonly used conductor materials for overhead lines are copper, aluminum, steel-cored aluminium, galvanised steel and cadmium copper. The choice of a particular material will depend upon the cost, the required electrical and mechanical properties and the local conditions.

Copper: Copper is an ideal material for overhead lines owing to its high electrical conductivity and greater tensile strength. It is always used in the hard drawn form as stranded conductor. Copper has high current density. However, due to its higher cost and non-availability, it is rarely used for these purposes. Now-a-days the trend is to use aluminium in place of copper.

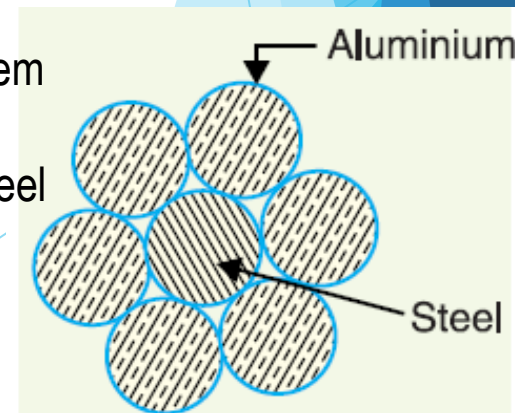
Conductor Materials

Commonly Used Conductor Materials:

Aluminum: Aluminium is cheap and light as compared to copper but it has much smaller conductivity and tensile strength. Aluminum replaces copper, for same resistance, because:

- Larger diameter - [about 1.26 times] (Resistance varies inversely with the diameter. As the diameter gets larger so the resistance of the wire gets smaller and get more space to carry current through)
- Lower voltage gradient at the surface of the conductor
- Less tendency to develop corona
- Lower cost - (About 6 times cheaper)
- Lighter than copper- (Twice lighter than copper), therefore specific gravity of Al (2.71 gm/cc) is lower than that of Cu (8.9 gm/cc). This reduces the cost of tower also, as it need not to be made so strong as that of Cu conductor

Steel-Cored Aluminum: Due to low tensile strength, aluminium conductors produce greater sag. This prohibits their use for larger spans and makes them unsuitable for long distance transmission. In order to increase the tensile strength, the aluminium conductor is reinforced with a core of galvanised steel wires. The composite conductor thus obtained is known as steel cored aluminium and is abbreviated as A.C.S.R. (aluminium conductor steel reinforced). Steel-cored aluminium conductor consists of central core of galvanised steel wires surrounded by a number of aluminium strands.



Conductor Materials

Commonly Used Conductor Materials:

Galvanised steel: Steel has very high tensile strength. Therefore, galvanised steel conductors can be used for extremely long spans or for short line sections exposed to abnormally high stresses due to climatic conditions. They have been found very suitable in rural areas where cost is the main consideration. Due to poor conductivity and high resistance of steel, such conductors are not suitable for transmitting large power over a long distance.

Cadmium Copper: An addition of 1% to 2% Cadmium to Copper increases the tensile strength by about 50%. This type of conductor could be used for exceptionally long span transmission lines. However, due to high cost of Cadmium they will be economical only for small cross-section transmission lines.

Line Supports

Types

In general, line supports should have the following properties:

- High mechanical strength to withstand the weight of conductors and wind loads etc.
- Light in weight without the loss of mechanical strength.
- Cheap in cost and economic to maintain.
- Longer life.
- Easy accessibility of conductors for maintenance.

Wooden Poles:

Economical and widely used for lower voltage distribution lines, especially in rural areas.

Steel Poles:

Offer greater strength and longer spans than wooden poles, often used in urban distribution networks.

Reinforced Concrete Poles:

Suitable for areas with high corrosion or environmental exposure, providing good durability.

Steel Lattice Towers:

Used for high-voltage transmission lines over long distances, known for their high strength and ability to handle long spans.

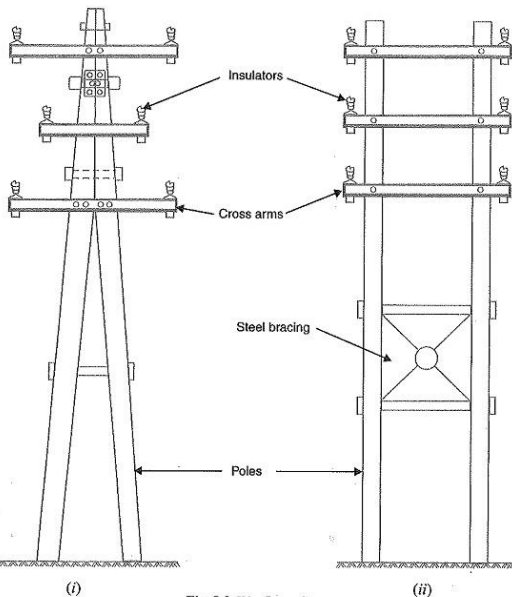


Fig. 8.2 Wooden poles

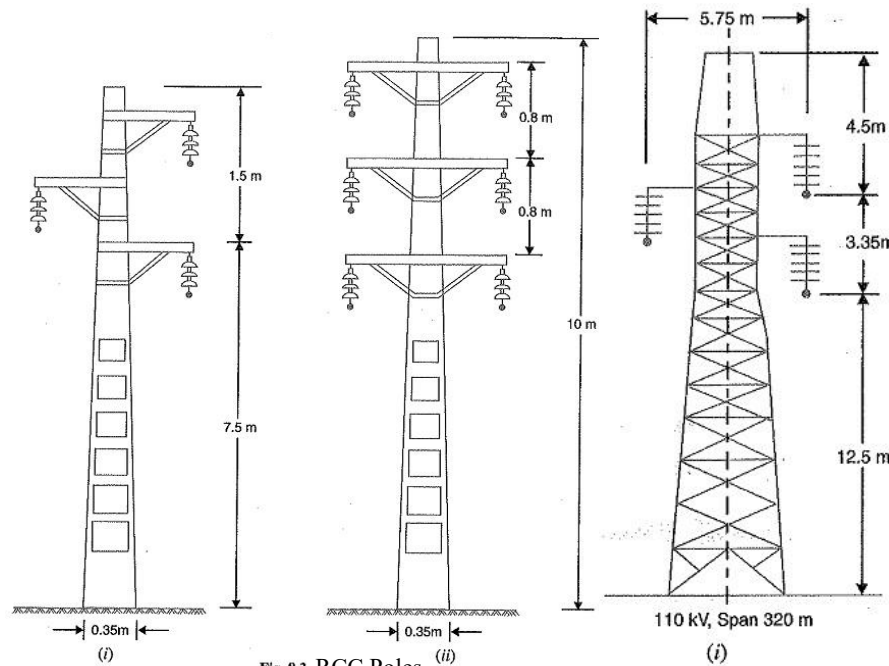


Fig. 8.3 RCC Poles

Fig. 8.4 Steel Towers

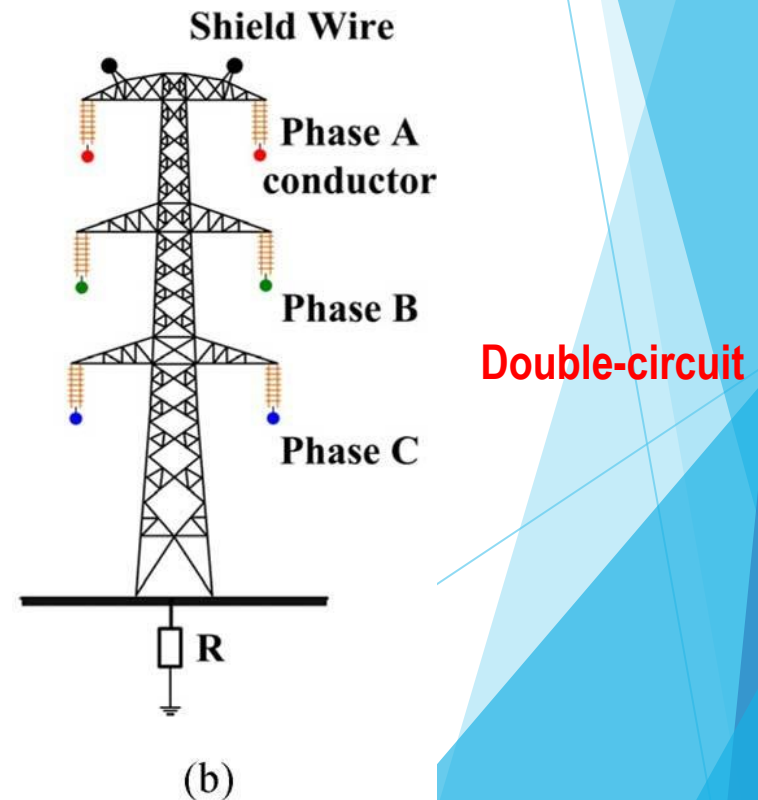
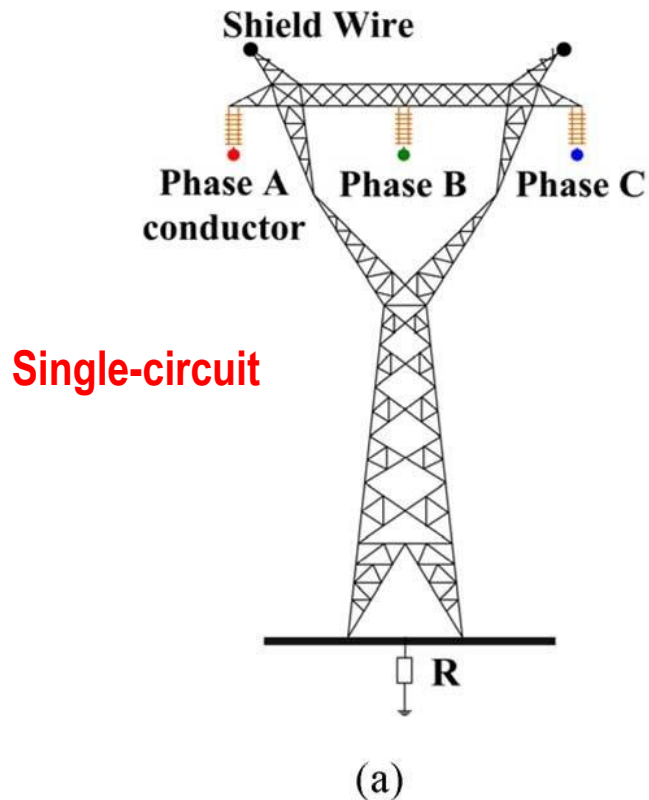
Line Supports

Properties	Wooden poles	Steel poles	Concrete poles
Mechanical strength & span	Less mechanical strength and therefore relatively shorter spans (50 m)	Greater mechanical strength, and permit longer spans to be used than wooden poles.	Greater mechanical strength and permit longer spans than steel poles.
Cost and insulating properties	Cheap, easily available, provide insulating properties	Need to be galvanized or painted periodically to prolong its life and prevent them from corrosion and hence maintenance cost is high.	Require little maintenance and have good insulating properties.
Voltage level	Suitable for lines of moderate cross sectional area and cannot be used for voltages higher than 20 kV.	Used for higher voltages.	Used for higher voltages.
Life	Tendency to rot below the ground level. Therefore comparatively smaller life (20-25 years) and require periodical inspection.	Longer life of at least 30 years.	Longer life.
Application	Can be used for distribution purposes in rural areas.	Can be used for distribution purposes in city areas. Can be used as the substitute for wooden poles.	Use of concrete poles is suitable and preferred in the coastal zones.

Line Supports

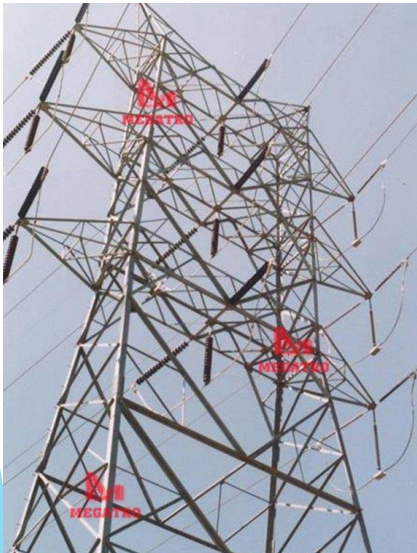
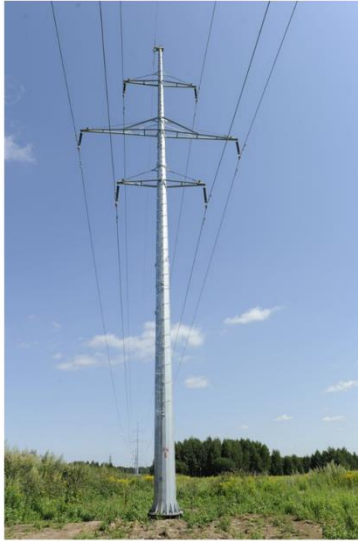
A **single-circuit** alternating current (AC) transmission line has three phases. At low voltages, a phase usually consists of one conductor. At high voltages (over 200 kV), a phase can consist of multiple conductors (bundled) separated by short spacers. A **double-circuit** AC transmission line has two sets of three phases.

Double-circuit structures typically hold the conductors in a vertical or stacked configuration, whereas single-circuit structures typically hold the conductors horizontally. Due to the vertical configuration of the conductors, double-circuit structures are taller than single-circuit structures. On lower voltage lines, structures sometimes carry more than two circuits.



Line Supports

Can you identify the circuit structures?



Insulators

The overhead line conductors should be supported on the poles or towers in such a way that currents from conductors do not flow to earth through supports i.e. line conductors must be properly insulated from supports. In general, insulators should have the following desirable properties:

- (i) **High mechanical strength** in order to withstand conductor load, wind load etc.
- (ii) **High electrical resistance** of insulator material in order to avoid leakage currents to earth.
- (iii) **High relative permittivity** of insulator material in order that dielectric strength is high.
- (iv) The insulator material should be **non-porous**, free from impurities and cracks otherwise the permittivity will be lowered.
- (v) **High ratio of puncture strength to flashover.**



The most commonly used material for insulators of overhead line is **porcelain** but **glass**, **steatite** and special composition materials are also used to a limited extent. The choice between glass and porcelain insulators often depends on the specific application, environmental conditions, and budget considerations.

Advantages of Glass Insulators

- The **skin of glass insulator is more restraint to damage** by chipping and by power arc than porcelain.
- It is **easier to detect a damaged glass** insulator than a porcelain unit.
- **Dielectric strength of glass is much higher** than that of porcelain.
- Glass insulator has a **lower coefficient of thermal expansion** than porcelain thus minimizing the strains due to temperature change.
- Glass material is transparent to light thereby heating up only slightly when exposed to light.
- Glass insulator **is mechanically stronger than porcelain** when under compression; in tension it has about the same strength as porcelain.

Disadvantages of Glass Insulators

- **Moisture more readily condenses** on the surface of glass insulator.
- Large size of mass of the material combined with the irregular shape may result in internal strain after cooling.

Insulators

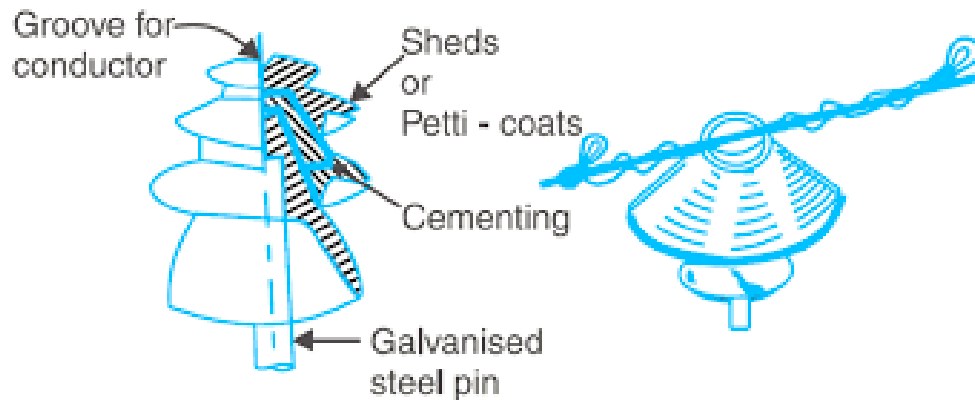
Types of Insulators

Pin type insulators

- **Pin type insulators** consists of a single or multiple shells.
- Multiple shells are provided in order to withstand more voltage stress.
- They are secured on the cross arms of the pole to carry power lines.
- There is a groove on the upper end of a pin insulator for housing the conductor. Conductor wire is passed through this groove and secured by binding with the same wire as of conductor.
- **Pin insulators** are used in distribution systems up to 33 kV voltage level. Beyond operating voltage of 33 kV, the pin type insulators become too bulky and hence uneconomical.

The most commonly used types include:

- ➡ **Pin type** insulators
- ➡ **Suspension type** insulators
- ➡ **Strain** insulators
- ➡ **Shackle** insulators



Insulators

Suspension type insulators

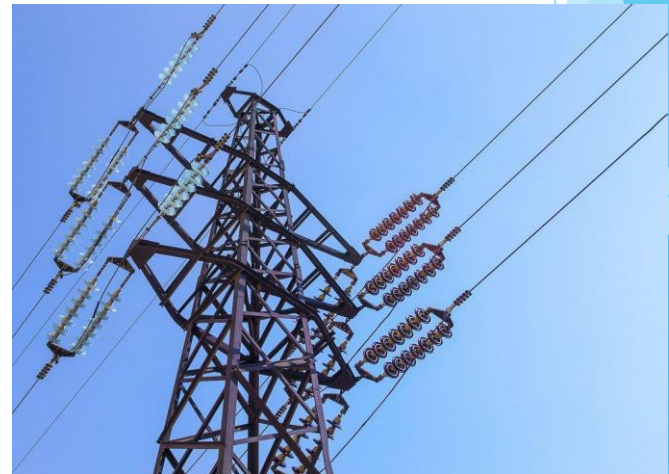
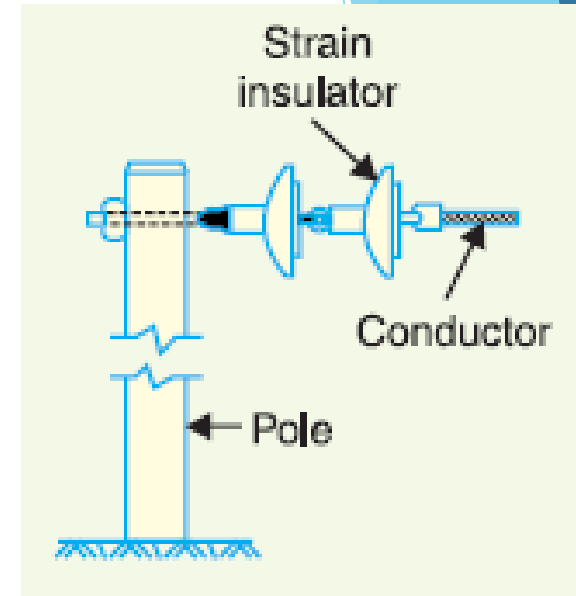
- In **suspension insulator**, numbers of **insulators** are connected in series to form a string.
- Each **insulator** of a **suspension** string is called **disc insulator** because of their disc like shape. The number of discs in series would obviously depend upon the working voltage. Each unit or disc is designed for low voltage, say 11 kV. For instance, if the working voltage is 66 kV, then six discs in series will be provided on the string.
- The line conductor is carried by the bottom most **insulator**.
- The suspension type insulators are generally **used with steel towers**.
- Suspension type insulators are cheaper than pin type insulators for voltages **beyond 33 kV**.
- If any one **disc is damaged**, the whole string does not become useless because the damaged disc can be replaced by the good one.
- The suspension arrangement provides greater **flexibility to the line**. The connection at the cross arm is such that insulator string is free to swing in any direction and can take up the position where mechanical stresses are minimum.



Insulators

Strain insulators

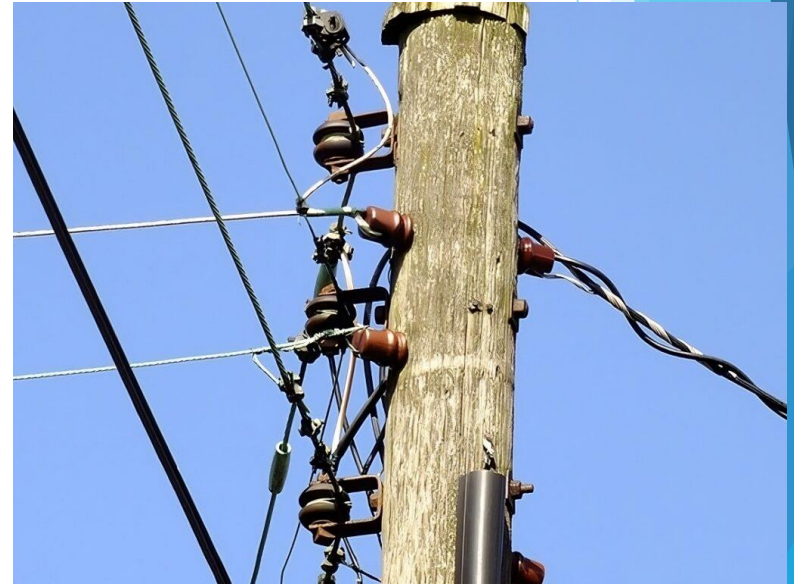
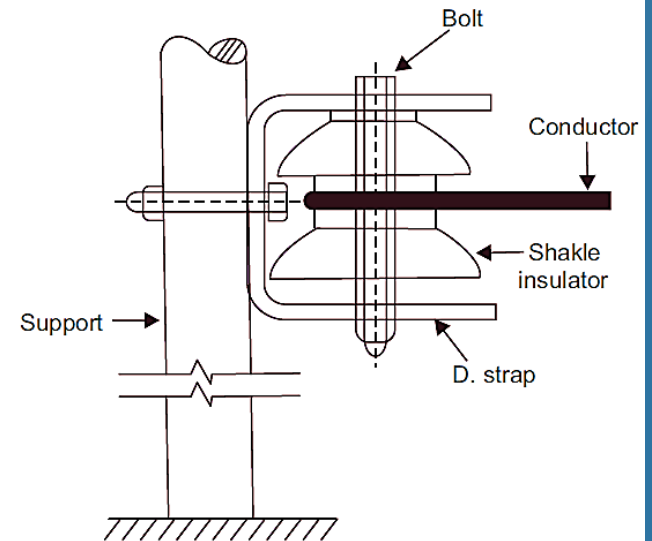
- At a dead end of a transmission line or at a corner or sharp curve, the transmission line is subjected to a great tensile load (strain). In order to sustain this great tension, strain insulators are used at dead ends.
- When the tension in lines is exceedingly high, as at long river spans, two or more strings are used in parallel.



Insulators

Shackle insulators

- In early days, the shackle insulators were used as strain insulators. But now a days, they are frequently used for low voltage distribution lines. Such insulators can be used either in a horizontal position or in a vertical position.
- They can be directly fixed to the pole with a bolt or to the cross arm. The conductor in the groove is fixed with a soft binding wire.



Insulators

Causes of Insulator Failure

- Insulators are required to withstand both mechanical and electrical stresses.
- The electrical breakdown of the insulator can occur either by *flash-over* or *puncture*.

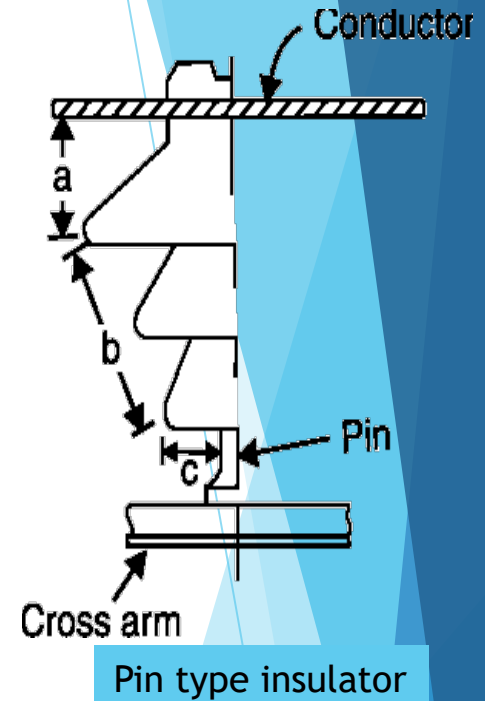
In **flashover**, an arc occurs between the line conductor and insulator pin (i.e., earth). Dirt, pollution, salt, and particularly water on the surface of a high voltage insulator can create a conductive path across it, causing leakage currents and flashovers. In case of flash-over, the insulator will continue to act in its proper capacity unless extreme heat produced by the arc destroys the insulator.

In case of **puncture**, the discharge occurs from conductor to pin through the body of the insulator due to excessive voltage level. When such breakdown is involved, the insulator is permanently destroyed due to excessive heat. In practice, sufficient thickness of porcelain is provided in the insulator to avoid puncture by the line voltage.

The ratio of puncture strength to flashover voltage is known as safety factor i.e.,

$$\text{Safety factor of insulator} = \frac{\text{Puncture strength}}{\text{Flash-over voltage}}$$

It is desirable that the value of safety factor is high so that flash-over (continuous process) takes place before the insulator gets punctured (ultimate failure). For pin type insulators, the value of safety factor is about 10.



Insulators

Causes of Insulator Failure

Types of Flashover Voltage

Power Frequency Flashover Voltage:

The RMS value of an AC voltage (at 50 or 60 Hz) that causes a flashover. This is usually measured under two conditions:

Dry Flashover Voltage: The voltage at which air surrounding the insulator breaks down under strictly dry laboratory conditions.

Wet Flashover Voltage: The voltage at which flashover occurs when the insulator is artificially sprayed with water at a 45-degree angle to simulate heavy rain.

Impulse Flashover Voltage:

The peak voltage measured during rapid, high-magnitude transient events such as lightning strikes or switching surges.

This test is to ensure that the insulator is capable of sustaining high voltage surges caused by lightning. The insulator under test is mounted in the same manner as in above tests. An impulse voltage generator which generates a very high voltage at a frequency of several hundred kilohertz is connected to the insulator. This voltage is applied to the insulator and spark-over voltage is noted.

$$\text{Safety Factor} = \frac{\text{Puncture Voltage}}{\text{Power Frequency Dry Flashover Voltage}}$$

Insulators

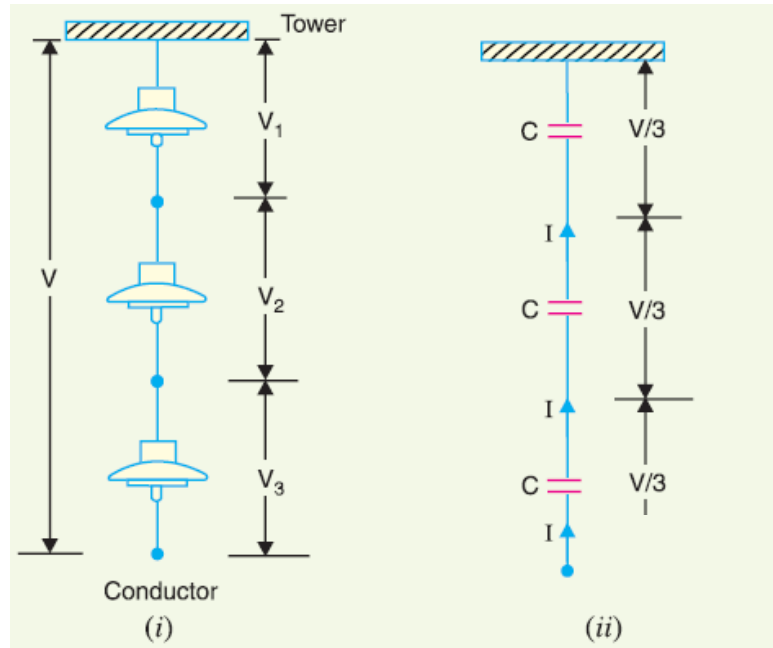
Causes of Insulator Failure

Types of Flashover Voltage

Characteristic	Dry Flashover	Wet Flashover	Impulse Flashover
Voltage Type	Power-frequency (AC/DC), slow rise	Power-frequency (AC/DC), slow rise	High-frequency transient surge (Lightning/Switching)
Environmental Condition	Clean and completely dry	Wet / Heavy rain	Can be dry or wet (usually measured dry)
Arc Path	Follows insulator profile or striking distance	Bridges across wet sheds; takes shortened path	Shortest straight-line air path between terminals
Relative Voltage Level	Medium (Baseline)	Lowest (Greatly reduced by moisture)	Highest (Due to rapid voltage rise time)
Primary Design Countermeasure	Adequate leakage (creepage) distance	Adding sheds/petticoats to create dry zones	Increasing overall striking distance and using surge arresters

Insulators

Causes of Insulator Failure



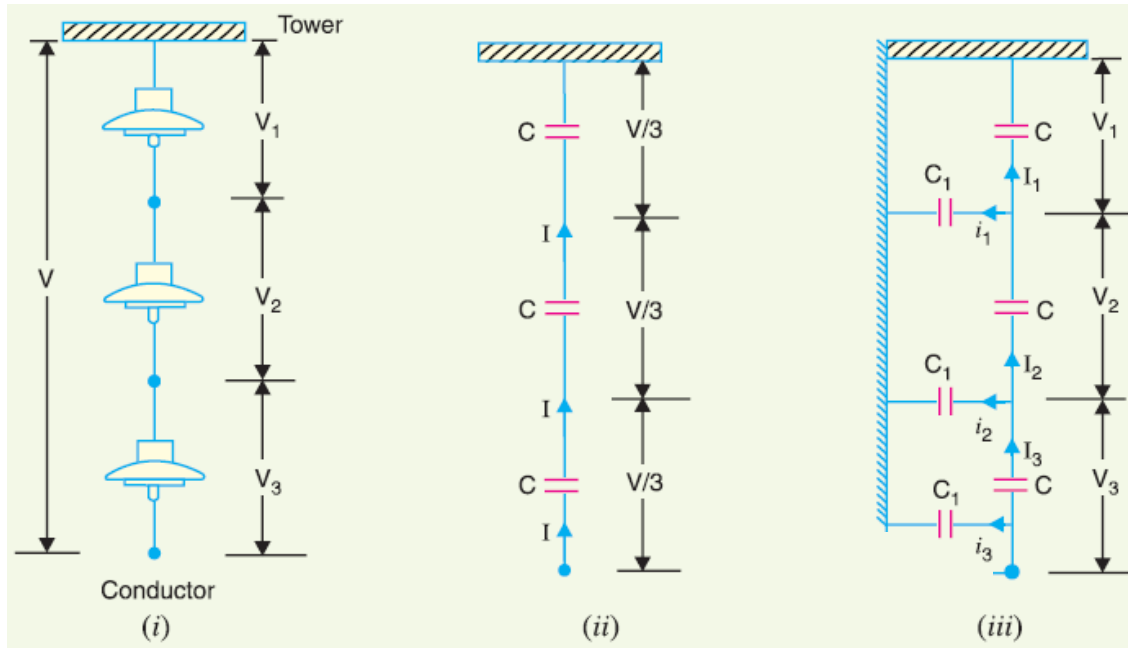
Suspension type insulator

Potential Distribution over Suspension Insulator String

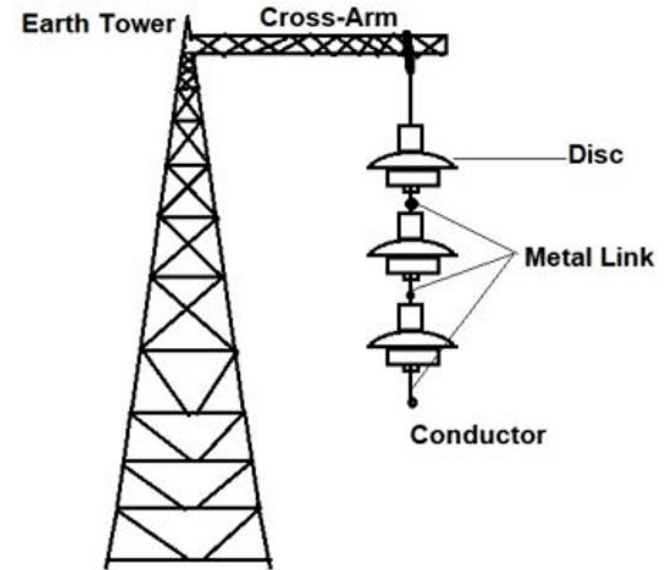
String of suspension insulators consists of a number of porcelain discs connected in series through metallic links. The porcelain portion of each disc is in between two metal links as shown in Fig. (i). Therefore, each disc forms a capacitor C . This is known as mutual capacitance or self-capacitance. If there were mutual capacitance alone, then charging current would have been the same through all the discs and consequently voltage across each unit would have been the same i.e., $V/3$ as shown in Fig. (ii).

Insulators

Causes of Insulator Failure



Suspension type insulator



However, in actual practice, capacitance also exists between metal fitting of each disc and tower or earth. This is known as shunt capacitance C_1 . Due to shunt capacitance, charging current is not the same through all the discs of the string [See Fig. (iii)]. Therefore, voltage across each disc will be different. Obviously, the disc nearest to the line conductor will have the maximum voltage because the charging current has its maximum value at the disc nearest to the conductor. V_3 will be much more than V_2 or V_1 .

Hence, the disc nearest to the conductor is under maximum electrical stress and is likely to be punctured. Therefore, means must be provided to equalize the potential across each unit.

Insulators

Causes of Insulator Failure

Mathematical expression of Potential Distribution over Suspension Insulator String

Let us suppose that self capacitance of each disc is C . Let us further assume that shunt capacitance C_1 is some fraction K of self-capacitance *i.e.*, $C_1 = KC$. Starting from the cross-arm or tower, the voltage across each unit is V_1, V_2 and V_3 respectively as shown.

Applying Kirchhoff's current law to node A, we get,

$$I_2 = I_1 + i_1$$

or

$$V_2 \omega C^* = V_1 \omega C + V_1 \omega C_1$$

or

$$V_2 \omega C = V_1 \omega C + V_1 \omega K C$$

$\therefore$

$$V_2 = V_1 (1 + K)$$

...(i)

Applying Kirchhoff's current law to node B, we get,

$$I_3 = I_2 + i_2$$

or

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) \omega C_1$$

or

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) \omega K C$$

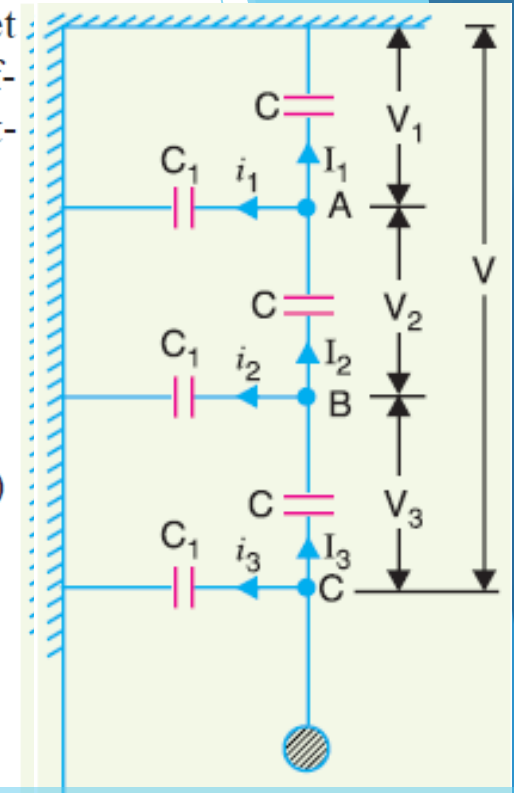
or

$$\begin{aligned} V_3 &= V_2 + (V_1 + V_2)K \\ &= KV_1 + V_2 (1 + K) \\ &= KV_1 + V_1 (1 + K)^2 \\ &= V_1 [K + (1 + K)^2] \end{aligned}$$

$\therefore$

$$V_3 = V_1 [1 + 3K + K^2]$$

(ii)



Voltage across 2nd shunt capacitance C_1 from top = $V_1 + V_2$. It is because one point of it is connected to B and the other point to the tower.

Insulators

Causes of Insulator Failure

Mathematical expression of Potential Distribution over Suspension Insulator String

Voltage between conductor and earth (*i.e.*, tower) is

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ &= V_1 + V_1(1 + K) + V_1(1 + 3K + K^2) \\ &= V_1(3 + 4K + K^2) \end{aligned}$$

$$\therefore V = V_1(1 + K)(3 + K)$$

From expressions (i), (ii) and (iii), we get,

$$\frac{V_1}{1} = \frac{V_2}{1 + K} = \frac{V_3}{1 + 3K + K^2} = \frac{V}{(1 + K)(3 + K)}$$

$$\therefore \text{Voltage across top unit, } V_1 = \frac{V}{(1 + K)(3 + K)}$$

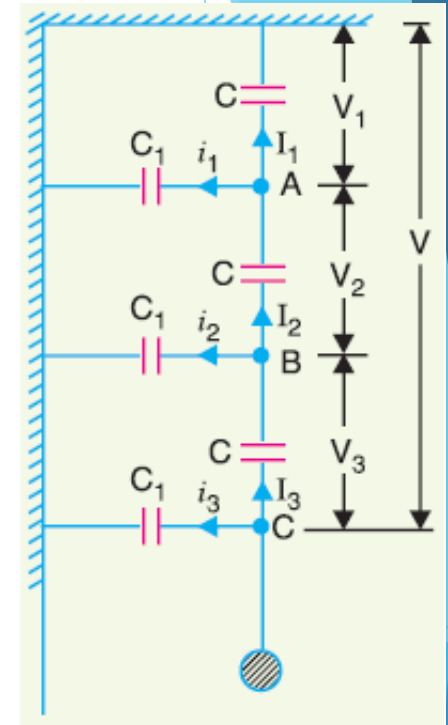
$$\text{Voltage across second unit from top, } V_2 = V_1(1 + K)$$

$$\text{Voltage across third unit from top, } V_3 = V_1(1 + 3K + K^2)$$

If $K = 0.2$ (Say), then from exp. (iv), we get, $V_2 = 1.2 V_1$ and $V_3 = 1.64 V_1$. This clearly shows that disc nearest to the conductor has maximum voltage across it; the voltage across other discs decreasing progressively as the cross-arm is approached.

What if $k=0$?

The greater the value of $K (= C_1/C)$, the more non-uniform is the potential across the discs and lesser is the string efficiency.



Insulators

String Efficiency

The unequal potential distribution across discs of suspension type insulator is undesirable and is usually expressed in terms of String Efficiency.

$$\text{String Efficiency} = \frac{\text{Voltage across the string}}{n \times \text{Voltage across disc nearest to conductor}}$$

$$\begin{aligned}\% \text{age String efficiency} &= \frac{\text{Voltage across string}}{n \times \text{Voltage across disc nearest to conductor}} \times 100 \\ (\text{for } n=3) &= \frac{V}{3 \times V_3} \times 100\end{aligned}$$

n = number of discs in the string.

The greater the string efficiency, the more uniform the voltage distribution; 100% string efficiency is an ideal case;

Methods of Improving String Efficiency

- By Using Longer Cross Arm
- By Grading The Insulators
- By Using Guard Ring

Insulators

Methods of Improving String Efficiency

By using longer cross arm:

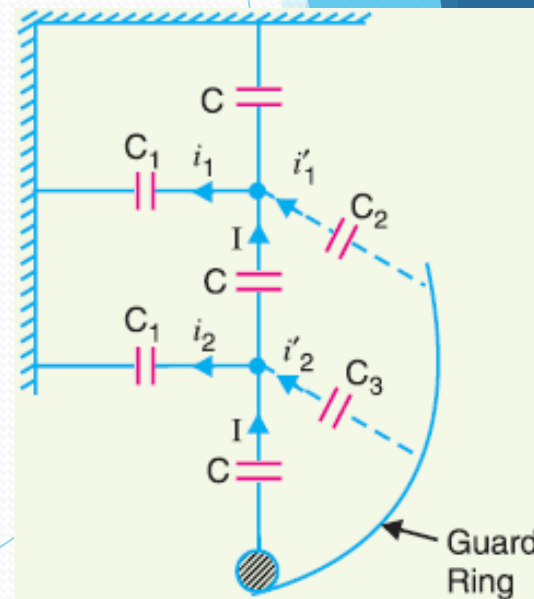
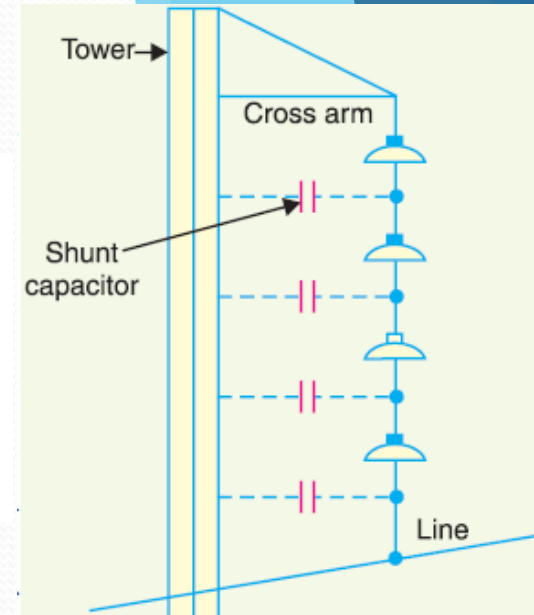
- String efficiency depends upon k .
- The least the value of k the more uniform is the voltage distribution.
- k can be decreased by reducing shunt capacitance. Shunt capacitance can be decreased by increasing longer cross arm. There is a limit of using longer cross arm (In practice, $k=0.1$ is the limit).

By grading the insulators:

- Insulators of different dimensions are so chosen that each has a different capacitance.
- Insulators are capacitance graded.
- Top unit has the minimum capacitance (since voltage is inversely proportional to the capacitance, $V=Q/C$).
- This method has the disadvantage that a large numbers of different-sized insulators are required.

By using a guard ring:

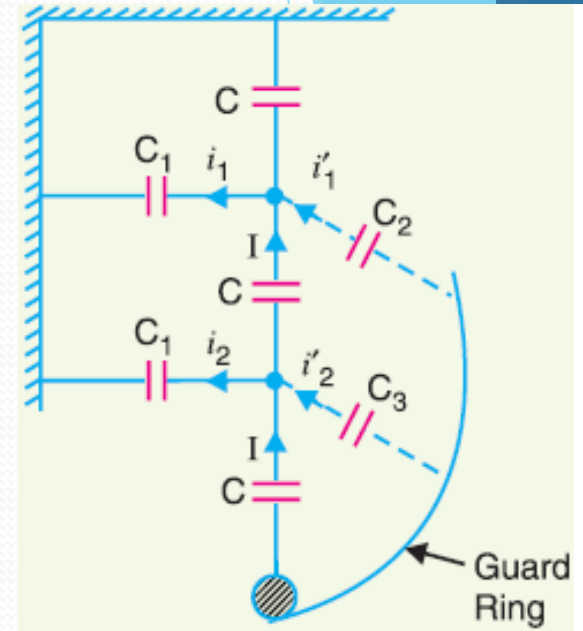
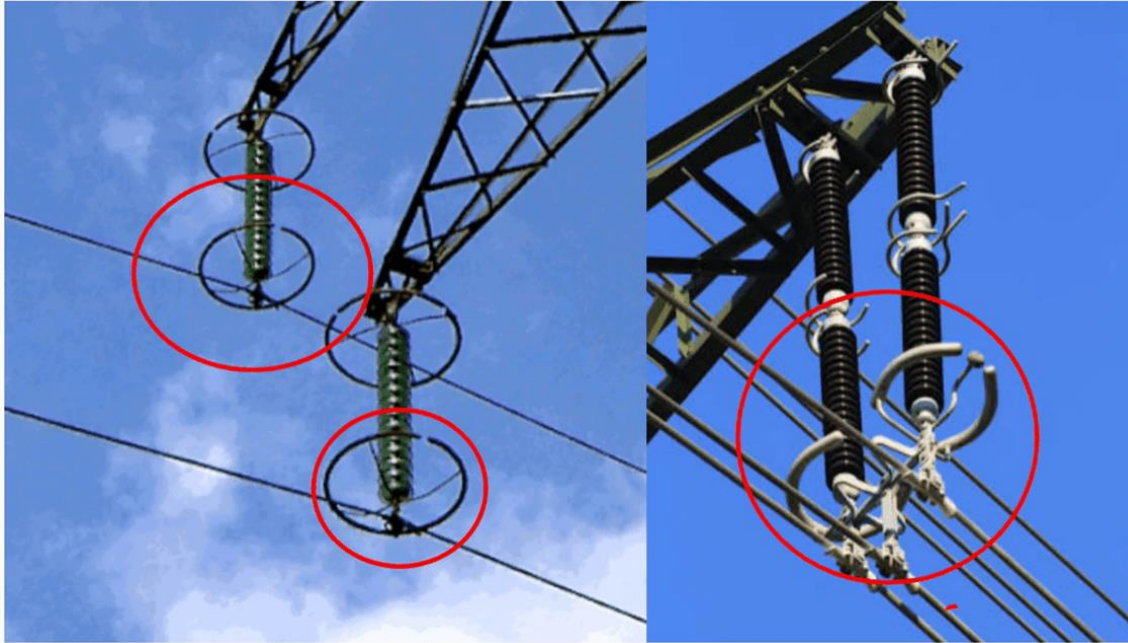
- Guard ring introduces capacitance between metal fittings and the line conductor.
- The guard ring is contoured in such a way that shunt capacitance currents i_1, i_2 etc. are respectively equal to i'_1, i'_2 etc.
- The result is that the same charging current I flows through each unit of string.



Insulators

Methods of Improving String Efficiency

By using a guard ring:



Guard rings can also help prevent flashover (an arc discharge) by reducing the electric field concentration near the bottom insulator and providing a path for current to flow during overvoltages.

Insulators

Calculating number of insulator discs

Calculating the number of discs in a suspension insulator string involves dividing the **line-to-line voltage** by the **rated voltage per disc** (usually 11 kV), multiplying by $\sqrt{3}$, and then adding a **mechanical safety factor**

Calculation Formula

The basic formula to find the number of discs (N) is:

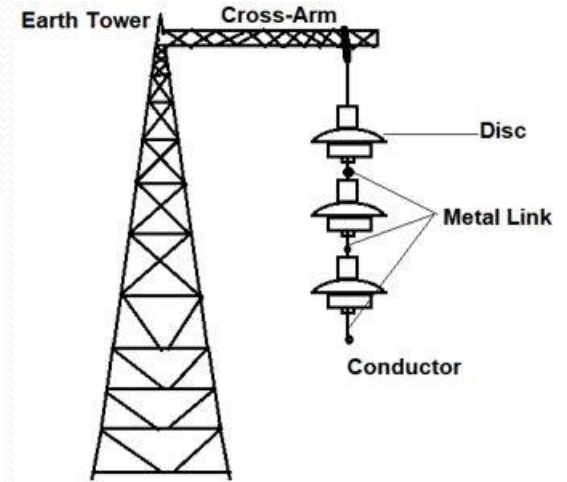
$$N = \frac{V_L}{\sqrt{3} \times V_{disc}} + S_f$$

Where:

- V_L = Line voltage of the transmission system (in kV)
- V_{disc} = Normal voltage rating of a single disc (typically 11 kV)
- S_f = Mechanical safety factor (usually 1 to 2, depending on the design)

Suspension Towers ($S_f = 1$): These towers are on a straight run. The insulator string hangs vertically, holding only the downward weight of the wire. The mechanical stress is low, so usually only 1 extra disc is added.

Tension / Dead-End Towers ($S_f = 2$): These towers are located at sharp turns, rivers, or terminal points. The insulator string hangs horizontally and pulls against the massive physical tension of the entire line span. Because a failure here is catastrophic, 2 extra discs are added for higher mechanical strength.



Why is it called a "Mechanical" Safety Factor?

The extra value S_f added to the end of the equation is often called a mechanical safety factor because its main job is physical reliability:

String Integrity: If one disc physically shatters or punctures due to a mechanical impact (like a bird strike or lightning stress), the remaining discs must have enough insulation to prevent the entire line from dropping.

Tension and Wind: Dead-end towers experience immense mechanical pulling forces. Extra discs are added here purely to handle the physical structural load, not because the voltage changed.

Example 8.1. In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self-capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) string efficiency.

Solution. Fig. 8.14. shows the equivalent circuit of string insulators. Let V_1 , V_2 and V_3 be the voltage across top, middle and bottom unit respectively. If C is the self-capacitance of each unit, then KC will be the shunt capacitance.

$$K = \frac{\text{Shunt Capacitance}}{\text{Self-capacitance}} = 0.11$$

$$\text{Voltage across string, } V = 33/\sqrt{3} = 19.05 \text{ kV}$$

At Junction A

$$I_2 = I_1 + i_1$$

or

$$V_2 \omega C = V_1 \omega C + V_1 K \omega C$$

or

$$V_2 = V_1 (1 + K) = V_1 (1 + 0.11)$$

or

$$V_2 = 1.11 V_1 \quad \dots(i)$$

At Junction B

$$I_3 = I_2 + i_2$$

or

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) K \omega C$$

or

$$\begin{aligned} V_3 &= V_2 + (V_1 + V_2) K \\ &= 1.11 V_1 + (V_1 + 1.11 V_1) 0.11 \end{aligned}$$

$\therefore$

$$V_3 = 1.342 V_1$$

(i) Voltage across the whole string is

$$V = V_1 + V_2 + V_3 = V_1 + 1.11 V_1 + 1.342 V_1 = 3.452 V_1$$

or

$$19.05 = 3.452 V_1$$

$$\therefore \text{Voltage across top unit, } V_1 = 19.05/3.452 = \mathbf{5.52 \text{ kV}}$$

$$\text{Voltage across middle unit, } V_2 = 1.11 V_1 = 1.11 \times 5.52 = \mathbf{6.13 \text{ kV}}$$

$$\text{Voltage across bottom unit, } V_3 = 1.342 V_1 = 1.342 \times 5.52 = \mathbf{7.4 \text{ kV}}$$

$$(ii) \quad \text{String efficiency} = \frac{\text{Voltage across string}}{\text{No. of insulators} \times V_3} \times 100 = \frac{19.05}{3 \times 7.4} \times 100 = \mathbf{85.8\%}$$

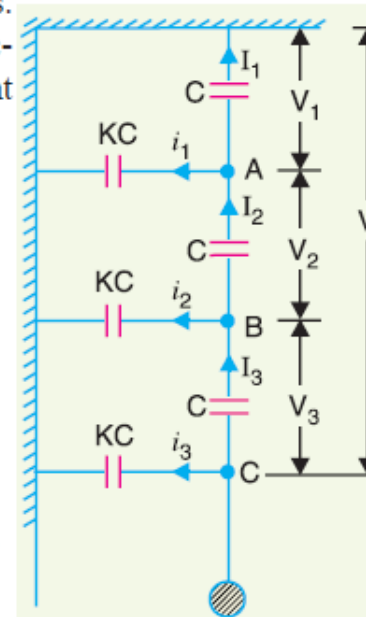


Fig. 8.14

Practice Problem

Example 8.3. Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is 1/8th of the capacitance of the insulator itself. Also find the string efficiency.

Practice Problem

Example 8.3. Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is 1/8th of the capacitance of the insulator itself. Also find the string efficiency.

Solution. Fig. 8.15 shows the equivalent circuit of string insulators. If C is the self capacitance of each unit, then KC will be the shunt capacitance where $K = 1/8 = 0.125$.

Voltage across line unit, $V_3 = 17.5$ kV

At Junction A

$$I_2 = I_1 + i_1$$

$$V_2 \omega C = V_1 \omega C + V_1 K \omega C$$

or

$$V_2 = V_1 (1 + K) = V_1 (1 + 0.125)$$

$\therefore$

$$V_2 = 1.125 V_1$$

At Junction B

$$I_3 = I_2 + i_2$$

or

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) K \omega C$$

or

$$V_3 = V_2 + (V_1 + V_2) K$$

$$= 1.125 V_1 + (V_1 + 1.125 V_1) \times 0.125$$

$\therefore$

$$V_3 = 1.39 V_1$$

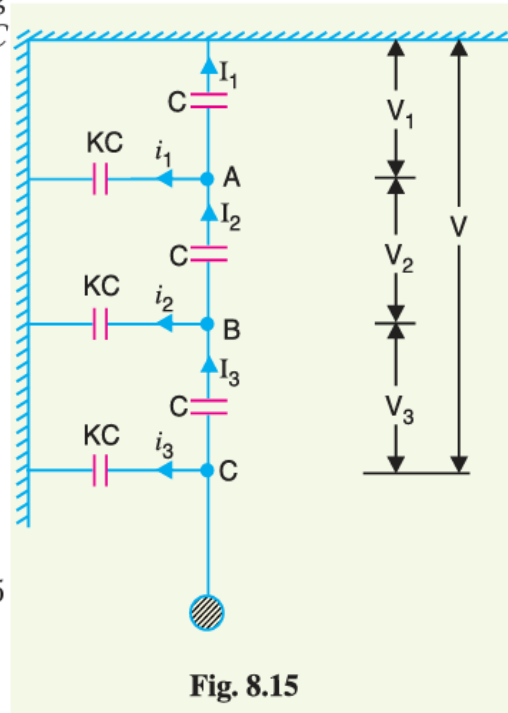
Voltage across top unit, $V_1 = V_3 / 1.39 = 17.5 / 1.39$
 $= 12.59$ kV

Voltage across middle unit, $V_2 = 1.125 V_1 = 1.125 \times 12.59 = 14.16$ kV

$\therefore$ Voltage between line and earth (i.e., line to neutral)

$$= V_1 + V_2 + V_3 = 12.59 + 14.16 + 17.5 = \mathbf{44.25 \text{ kV}}$$

$$\text{String efficiency} = \frac{44.25}{3 \times 17.5} \times 100 = \mathbf{84.28\%}$$



Practice Problem

Example 8.5. *An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency.*

Practice Problem

1. In a 3-phase overhead system, each line is suspended by a string of 3 insulators. The voltage across the top unit (*i.e.* near the tower) and middle unit are 10 kV and 11 kV respectively. Calculate (i) the ratio of shunt capacitance to self capacitance of each insulator, (ii) the string efficiency and (iii) line voltage.

[(i) 0.1 (ii) 86.76% (iii) 59 kV]

Practice Problem

1. In a 3-phase overhead system, each line is suspended by a string of 3 insulators. The voltage across the top unit (*i.e.* near the tower) and middle unit are 10 kV and 11 kV respectively. Calculate (i) the ratio of shunt capacitance to self capacitance of each insulator, (ii) the string efficiency and (iii) line voltage.

[(i) 0.1 (ii) 86.76% (iii) 59 kV]

1. Find the Ratio of Capacitances (K)

In a string of insulators, the relationship between the voltages of adjacent discs is given by the formula:

$$V_2 = V_1(1 + K)$$

Where:

- $V_1 = 10$ kV (Top unit)
- $V_2 = 11$ kV (Middle unit)
- $K = \frac{\text{Shunt Capacitance}}{\text{Self Capacitance}}$

Rearranging to solve for K :

$$11 = 10(1 + K)$$

$$K = 0.1$$

The ratio of shunt capacitance to self-capacitance is **0.1**.

2. Calculate Line Voltage

First, we need the voltage across the third (bottom) insulator (V_3). The formula for V_3 in a 3-disc string is:

$$V_3 = V_1(1 + 3K + K^2)$$

Plugging in our values:

$$V_3 = 10(1 + 3(0.1) + (0.1)^2)$$

$$V_3 = 10(1 + 0.3 + 0.01) = 10(1.31) = 13.1 \text{ kV}$$

Phase Voltage (V_{ph}):

The total voltage from the line to the tower (earth) is the sum of the voltages across the three discs:

$$V_{ph} = V_1 + V_2 + V_3$$

$$V_{ph} = 10 + 11 + 13.1 = 34.1 \text{ kV}$$

Line Voltage (V_L):

In a 3-phase system, the line voltage is $\sqrt{3}$ times the phase voltage:

$$V_L = \sqrt{3} \times 34.1 \approx 59.06 \text{ kV}$$

3. Calculate String Efficiency

String efficiency measures how uniformly the voltage is distributed. It is calculated as:

$$\text{String Efficiency} = \frac{\text{Total Voltage across string}}{n \times \text{Voltage across disc nearest to line}} \times 100$$

Where $n = 3$ (number of discs).

$$\text{Efficiency} = \frac{34.1}{3 \times 13.1} \times 100$$

$$\text{Efficiency} = \frac{34.1}{39.3} \times 100 \approx \mathbf{86.77\%}$$

Reference: Principles of Power System, V.K. Mehta & Rohit Mehta, 4th ed., S. Chand Publishing, 2005.
Chapter 8



QUESTIONS AND DISCUSSIONS

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